

Chapter-7 Electromagnetic Wave Propagation

How Maxwell Fixed Ampere's Law

↳ The differential form of basic laws of electricity and magnetism are as follows:

$$(i) \nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad (\text{Gauss's Law})$$

$$(ii) \nabla \cdot \vec{B} = 0$$

$$(iii) \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad (\text{Faraday's Law})$$

$$(iv) \nabla \times \vec{B} = \mu_0 \vec{J} \quad (\text{Ampere's Law})$$

Let us take the divergence of Eqⁿ (iii), i.e.

$$\nabla \cdot (\nabla \times \vec{E}) = \nabla \cdot \left(-\frac{\partial \vec{B}}{\partial t} \right) = -\frac{\partial}{\partial t} (\nabla \cdot \vec{B})$$

The left side is zero because divergence of curl is zero, the right side is zero by virtue of equation (ii).
Again,

Let us take divergence of Eqⁿ (iv) i.e.

$$\nabla \cdot (\nabla \times \vec{B}) = \mu_0 (\nabla \cdot \vec{J})$$

The left side must be zero, but the right side, in general, is not. For steady currents, the divergence of $\vec{J}$ is zero, but evidently when we go beyond magnetostatics Ampere's law cannot be right. So, Eqⁿ (iv) has to be modified.

Maxwell corrected Eq. (iv) by adding extra term to $\vec{J}$. let it be called $\vec{J}_d$, then

$$\nabla \times \vec{B} = \mu_0 (\vec{J} + \vec{J}_d)$$

$$\text{or, } \nabla \cdot (\nabla \times \vec{B}) = \mu_0 (\nabla \cdot \vec{J} + \nabla \cdot \vec{J}_d)$$

$$\text{or, } 0 = \mu_0 (\nabla \cdot \vec{J} + \nabla \cdot \vec{J}_d)$$

$$\begin{aligned} \text{or, } \nabla \cdot \vec{J}_d &= -\nabla \cdot \vec{J} = \frac{\partial \rho}{\partial t} = \frac{\partial (\epsilon_0 \nabla \cdot \vec{E})}{\partial t} \\ &= \nabla \cdot \epsilon_0 \frac{\partial \vec{E}}{\partial t} \end{aligned}$$

$$\therefore \vec{J}_d = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

~~Apart from~~ Thus the modified form of Ampere's law is

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

Apart from curing the defect in Ampere's law, Maxwell's term has a certain aesthetic appeal:

A changing electric field induces a magnetic field.

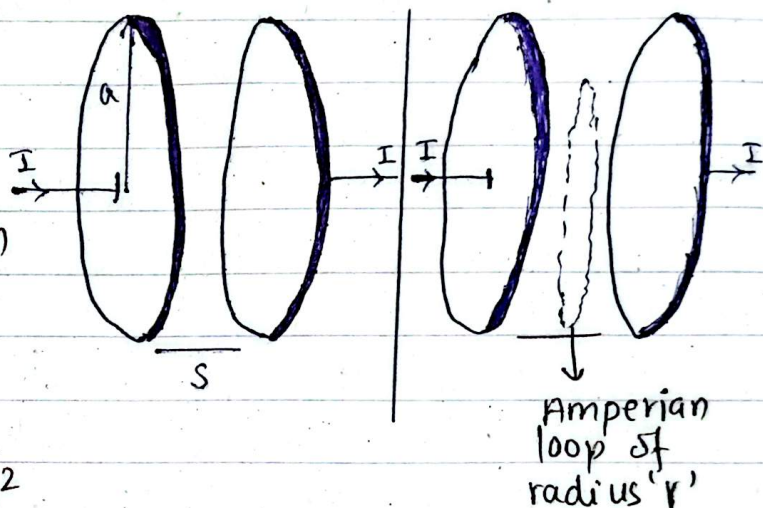
Maxwell called his extra term the

displacement current: $\vec{J}_d \equiv \epsilon_0 \frac{\partial \vec{E}}{\partial t}$

Q) A capacitor made from parallel circular plates, of radius a & separation s , is inserted into a long straight wire carrying current I . As the capacitor charges up, find the induced magnetic field midway between the plates, at a distance r ($r < a$) from the center.

Soln:

As the capacitor charges up, the magnitude of uniform electric field betⁿ the capacitor plates is.



$$E = \frac{\sigma}{\epsilon_0} = \frac{1}{\epsilon_0} \frac{Q}{A} = \frac{1}{\epsilon_0} \frac{Q}{\pi a^2}$$

The displacement current density is

$$J_d = \epsilon_0 \frac{\partial E}{\partial t} = \epsilon_0 \frac{\partial}{\partial t} \left(\frac{1}{\epsilon_0} \frac{Q}{\pi a^2} \right) = \frac{1}{\pi a^2} \frac{dQ}{dt} = \frac{I}{\pi a^2}$$

Ampere-Maxwell Law:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I + \mu_0 \epsilon_0 \int \left(\frac{\partial \vec{E}}{\partial t} \right) \cdot d\vec{a} = \mu_0 I + \mu_0 \int \vec{J}_d \cdot d\vec{a}$$

$$\Rightarrow B \times 2\pi r = 0 + \mu_0 J_d \pi r^2$$

$$\Rightarrow B = \frac{1}{2\pi r} \mu_0 \left[\frac{I}{\pi a^2} \right] \pi r^2$$

$$\therefore B = \frac{\mu_0 I}{2\pi a} r$$

Electromagnetic Waves in vacuum

The wave eqⁿ for $\vec{E}$ and $\vec{B}$.

In regions of space where there is no charge or current, Maxwell's equations are

$$\nabla \cdot \vec{E} = 0 \quad \text{--- (1)}$$

$$\nabla \cdot \vec{B} = 0 \quad \text{--- (2)}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \text{--- (3)}$$

$$\nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \text{--- (4)}$$

Taking curl of eqⁿ (3), we get,

$$\nabla \times (\nabla \times \vec{E}) = -\nabla \times \frac{\partial \vec{B}}{\partial t}$$

$$\text{or, } \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\frac{\partial(\nabla \times \vec{B})}{\partial t}$$

$$\therefore \nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

Again taking curl of eqⁿ (4), we get,

$$\nabla \times (\nabla \times \vec{B}) = \mu_0 \epsilon_0 \nabla \times \frac{\partial \vec{E}}{\partial t}$$

$$\text{or, } \nabla(\nabla \cdot \vec{B}) - \nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial(\nabla \times \vec{E})}{\partial t}$$

$$\therefore \nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}$$

In vacuum, then, each cartesian component of $\vec{E}$ & $\vec{B}$ satisfies the three-dimensional wave eqⁿ,

$$\nabla^2 f = \frac{1}{v^2} \frac{\partial^2 f}{\partial t^2}$$

So Maxwell's eqⁿs imply that empty space supports the propagation of electromagnetic waves, travelling at a speed

$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \approx 3 \times 10^8 \text{ m/s}$$

This is velocity of electromagnetic waves.

Poynting's Theorem

The total energy stored in electromagnetic field is

$$U_{em} = \int_V \frac{1}{2} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) d\tau \quad \text{--- (1)}$$

Suppose we have some charge and current configuration which, at time t , produces fields $\vec{E}$ and $\vec{B}$ and In the next interval dt , the charges move around a bit.

According to the Lorentz force law, the work done, dw , by the electromagnetic forces acting on a charge dq is

$$\begin{aligned} dw &= \vec{F} \cdot d\vec{r} = dq (\vec{E} + \vec{v} \times \vec{B}) \cdot d\vec{r} \\ &= dq (\vec{E} + \vec{v} \times \vec{B}) \cdot \vec{v} dt \\ &= \vec{E} \cdot \vec{v} dq dt \end{aligned}$$

The rate of total work done on all the charges in a volume V :

$$\frac{dW}{dt} = \int_V \vec{E} \cdot \vec{J} d\tau \quad [\because dq = \rho d\tau \text{ \& } \rho \vec{v} = \vec{J}] \quad \text{--- (2)}$$

From Ampere-Maxwell Law,

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\Rightarrow \vec{J} = \frac{1}{\mu_0} (\nabla \times \vec{B}) - \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\text{Now, } \vec{E} \cdot \vec{J} = \vec{E} \cdot \frac{1}{\mu_0} (\nabla \times \vec{B}) - \epsilon_0 \vec{E} \cdot \frac{\partial \vec{E}}{\partial t}$$

$$= \frac{1}{\mu_0} [\vec{B} \cdot (\nabla \times \vec{E}) - \nabla \cdot (\vec{E} \times \vec{B})] - \epsilon_0 \vec{E} \cdot \frac{\partial \vec{E}}{\partial t}$$

$$= \frac{1}{\mu_0} \left[\vec{B} \cdot \left(-\frac{\partial \vec{B}}{\partial t} \right) - \nabla \cdot (\vec{E} \times \vec{B}) \right] - \epsilon_0 \vec{E} \cdot \frac{\partial \vec{E}}{\partial t}$$

$$= \frac{1}{\mu_0} \left(-\frac{1}{2} \frac{\partial B^2}{\partial t} \right) - \frac{1}{\mu_0} \nabla \cdot (\vec{E} \times \vec{B}) - \epsilon_0 \frac{1}{2} \frac{\partial E^2}{\partial t} \quad [\because \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \text{ Faraday's law}]$$

$$\therefore \vec{E} \cdot \vec{J} = -\frac{1}{2} \frac{\partial}{\partial t} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) - \frac{1}{\mu_0} \nabla \cdot (\vec{E} \times \vec{B}) \quad \text{--- (3)}$$

putting $\vec{E}, \vec{J}$ into eqⁿ (2), we get,

$$\begin{aligned}\frac{dW}{dt} &= \int_V \left[-\frac{1}{2} \frac{\partial}{\partial t} \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) - \frac{1}{\mu_0} \nabla \cdot (\vec{E} \times \vec{B}) \right] d\tau \\&= -\frac{1}{2} \frac{d}{dt} \int_V \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) d\tau - \frac{1}{\mu_0} \int_V \nabla \cdot (\vec{E} \times \vec{B}) d\tau \\&= -\frac{1}{2} \frac{d}{dt} \int_V \left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2 \right) d\tau - \frac{1}{\mu_0} \oint_S (\vec{E} \times \vec{B}) \cdot d\vec{a}\end{aligned}$$

$$\therefore \frac{dW}{dt} = -\frac{dU_{em}}{dt} - \oint_S \vec{S} \cdot d\vec{a} \quad \text{--- (4)} \quad \left[\because \vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B}) \right]$$

↓
Poynting vector

where S is the surface bounding V .

This is Poynting theorem; it is the work-energy theorem of electrodynamics.

Poynting's theorem says that "the work done on the charges by the electromagnetic field is equal to the decrease in energy stored in the field, less the energy which is flowed out through the surface."

Poynting's theorem:

$$\frac{dW}{dt} = - \frac{dU_{em}}{dt} - \oint_S \vec{S} \cdot d\vec{a}$$

$$\text{or, } \frac{d}{dt} \int_V u_{mech} d\tau = - \frac{d}{dt} \int_V u_{em} d\tau - \oint_S \vec{S} \cdot d\vec{a}$$

$$\left[\begin{array}{l} \because u_{mech} = \text{mechanical energy density} \\ u_{em} = \text{energy density of the fields} \end{array} \right]$$

$$\text{or, } \frac{d}{dt} \int_V (u_{mech} + u_{em}) d\tau = - \int_V (\nabla \cdot \vec{S}) d\tau$$

$$\therefore \frac{\partial}{\partial t} (u_{mech} + u_{em}) = - \nabla \cdot \vec{S}$$

This is the differential version of Poynting's theorem.

Poynting's Vector:

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$

The energy per unit time, per unit area, transported by the fields.

SI unit of $\vec{S}$ are $\text{Js}^{-1}\text{m}^{-2}$ or Wm^{-2} .